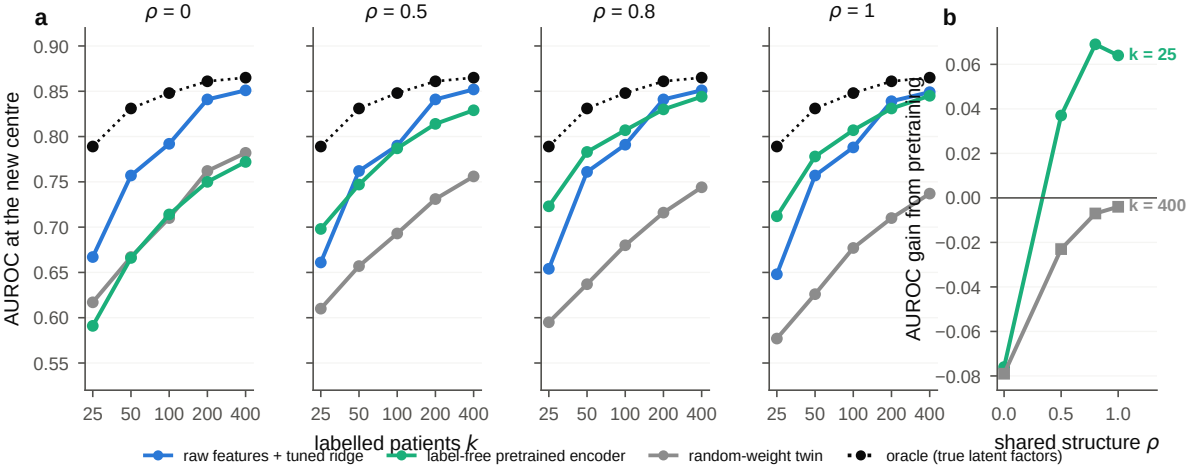
a
ρ = 0
ρ = 0.5
ρ = 0.8
ρ = 1
AUROC at the new centre
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
25
50
100
200
400
labelled patients k
b
AUROC gain from pretraining
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
k = 25
k = 400
0.0
0.5
1.0
shared structure ρ
raw features + tuned ridge
label-free pretrained encoder
random-weight twin
oracle (true latent factors)